

NanoGT Match Report: Achromatopsia

Disease: Achromatopsia (ORPHA:49382)

Primary gene: CNGB3

Gene CDS: 2427 bp

Inheritance: Autosomal recessive

Target tissues scored: retina

Gene selection note: this disease has multiple listed genes (CNGB3, CNGA3, GNAT2, PDE6C, PDE6H, ATF6); this report scores CNGB3 only.

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype; Vector does not naturally cover all annotated disease tissues: retina; Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Cone cyclic nucleotide-gated channel beta-subunit loss

Gene-addition compatibility: compatible

Preferred modality class: retinal gene addition

Evidence level/status: direct / source_linked_needs_review

Evidence summary: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable

Evidence source: OMIM CNGB3 gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Few catalog vectors cover the annotated disease tissue(s): AAV2.
- Endpoint readiness: retinal diseases often have measurable endpoints such as OCT, ERG, visual acuity, visual fields, or mobility testing, but genotype-specific progression still needs confirmation.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	CPCB-RPE1	AAV8	7.5/10	Medium	phase2/3
2	Luxturna	AAV2	7.1/10	Medium	approved
3	Skysona	LV	6.7/10	Medium	approved
4	Libmeldy	LV	6.3/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
5	Strimvelis	LV	6.3/10	Medium	approved

Match #1: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.70	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.48	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2427bp / cargo 4700bp (52% utilized)
- Precedent target match: retina
- Both membrane proteins
- Inheritance match (Autosomal recessive <-> AR)

- Pathway match: retinal_phototransduction
- Disease mechanism: loss of function — Cone cyclic nucleotide-gated channel beta-subunit loss
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable
- Mechanism source: OMIM CNGB3 gene entry (<https://omim.org/entry/605080>)
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CNGB3 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Vector does not naturally cover all annotated disease tissues: retina
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Luxturna

Precedent disease: Leber congenital amaurosis type 2

Vector: AAV2

Tissue target: retina/RPE

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway

Dimension	Score	Max	What it measures
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.14	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2427bp / cargo 4700bp (52% utilized)
- Vector tropism plus precedent target match: retina
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: retinal visual cycle
- Disease mechanism: loss of function — Cone cyclic nucleotide-gated channel beta-subunit loss
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable
- Mechanism source: OMIM CNGB3 gene entry (<https://omim.org/entry/605080>)
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CNGB3 standard subcellular localisation; nuclear

AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.67	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (retinal_phototransduction vs peroxisomal)
- Disease mechanism: loss of function — Cone cyclic nucleotide-gated channel beta-subunit loss
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable
- Mechanism source: OMIM CNGB3 gene entry (<https://omim.org/entry/605080>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CNGB3 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs leukodystrophy)
- Disease mechanism: loss of function — Cone cyclic nucleotide-gated channel beta-subunit loss
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable
- Mechanism source: OMIM CNGB3 gene entry (<https://omim.org/entry/605080>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1

- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CNGB3 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment

Dimension	Score	Max	What it measures
TOTAL (normalised)	6.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs immune_hematopoietic)
- Disease mechanism: loss of function — Cone cyclic nucleotide-gated channel beta-subunit loss
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Biallelic CNGB3 loss is compatible with photoreceptor-directed gene addition if target cells remain viable
- Mechanism source: OMIM CNGB3 gene entry (<https://omim.org/entry/605080>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CNGB3 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility